

Axiomatic Foundations of Scale-Dependent Geometric Field Theory

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We present the axiomatic foundations of Scale-Dependent Geometric Field Theory (SDGFT), a zero-parameter framework in which all physical couplings and mixing profiles emerge from two topological constants. These axioms, derived from the unique self-dual regular four-dimensional polytope (the 24-cell), are the Fibonacci-lattice conflict $\Delta = 5/24$ and the elementary lattice tension $\delta = 1/24$. We demonstrate how the golden ratio and the six-cone partition of the 3-sphere follow algebraically, setting the stage for a fully coordinate-independent description of spacetime and particle physics.

I. INTRODUCTION

We initiate a program to derive the fundamental parameters of particle physics and cosmology from pure geometry. By identifying the 24-cell as the base unit of discrete spacetime, we show that no free parameters are needed to describe the scale-dependent properties of physical systems.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\Delta = \frac{5}{24}, \quad \delta = \frac{1}{24} \quad (1)$$

$$\Delta + \delta = \frac{1}{4} = \sin^2 30^\circ \implies \theta_{\max} = 30^\circ \quad (2)$$

$$\frac{\Delta}{\delta} = 5 \implies \varphi = \frac{1 + \sqrt{5}}{2} \quad (3)$$

III. THE TOPOLOGICAL AXIOMS

The framework rests on two quantities: the lattice conflict Δ , reflecting Fibonacci self-similarity, and the tension δ . Their sum exactly corresponds to the opening angle of a 30-degree cone, partitioning the three-sphere into six symmetric sectors.

IV. CONCLUSION

We have established the topological foundation of SDGFT, providing a rigid framework that links microscopic gauge groups to macroscopic cosmic perturbations.

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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

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